



TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations
Revision Date: 03/10/2025

Version: 1.1

SECTION 1: IDENTIFICATION

Product Identifier

Product Form: Mixture
Product Name: TM 2000 RB
Product Code: AFCO 5652

Intended Use of the Product

Use of the Substance/Mixture: Neutral stone prevention detergent. For professional use only.

Name, Address, and Telephone of the Responsible Party

Company

AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329
www.afcocare.com

Emergency Telephone Number

Emergency Number : 1-800-424-9300 (CHEMTREC)

SECTION 2: HAZARDS IDENTIFICATION

Classification of the Substance or Mixture

Classification (GHS-US)

Eye Irrit. 2A H319
Aquatic Acute 3 H402
Aquatic Chronic 3 H412

Label Elements

GHS-US Labeling

Hazard Pictograms (GHS-US):



Signal Word (GHS-US)

: Warning.

Hazard Statements (GHS-US)

: H319 - Causes serious eye irritation
H402 - Harmful to aquatic life
H412 - Harmful to aquatic life with long lasting effects

Precautionary Statements (GHS-US)

: P264 - Wash hands, forearms, and other exposed areas thoroughly after handling.
P273 - Avoid release to the environment.
P280 - Wear protective glove, protective clothing, eye protection, face protection.
P305+P351+P338 - If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P337+P313 - If eye irritation persists: Get medical advice/attention.
P501 - Dispose of contents/container according to local, regional, national, territorial, provincial, and international regulations.

Other Hazards

Results of PBT and vPvB Assessment: This mixture does not contain any substances that are assessed to be a PBT or a vPvB.

Unknown Acute Toxicity (GHS-US): Not available.

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Substances: Not relevant (mixture).

Mixture

Name	Product identifier	% (w/w)	Classification (GHS-US)
2-phosphonobutane 1,2, 4 tricarboxylic acid, tetrasodium salt	(CAS No.) 66669-53-2	15-25	Acute Tox. 4 (Oral), H302 Eye Irrit. 2A, H319

TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Sodium salt of hydroxyethylidene-1,1, diphosphonic acid	(CAS No 29329-71-3)	1-5	Not classified.
Zinc sulfate heptahydrate	(CAS No.) 7447-20-0	<1.0	Acute Tox. 4 (Oral), H302 Eye Dam. 1, H318 Aquatic Acute 1 ,H400 Aquatic Chronic 1, H410

The exact percentage of composition are withheld as a trade secret [29 CFR 1910.1200].

Full text of H-phrases: see section 16.

SECTION 4: FIRST AID MEASURES

Description of First Aid Measures

General: Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

Inhalation: When symptoms occur: go into open air and ventilate suspected area. Remove to fresh air and keep at rest in a position comfortable for breathing. If breathing is irregular or stopped, immediately call a POISON CENTER or doctor/physician.

Skin Contact: Remove contaminated clothing. Drench affected area with water for several minutes. If irritation persists, immediately call a POISON CENTER or doctor/physician. Wash contaminated clothing before reuse.

Eye Contact: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If irritation occurs, immediately call a POISON CENTER or doctor/physician.

Ingestion: Rinse mouth. Do NOT induce vomiting. Immediately call a POISON CENTER or doctor/physician.

Most Important Symptoms and Effects Both Acute and Delayed

Symptoms and effects are not known to date.

Indication of Any Immediate Medical Attention and Special Treatment Needed

None.

SECTION 5: FIRE-FIGHTING MEASURES

Extinguishing Media

Suitable Extinguishing Media: Water spray, fog, carbon dioxide, foam, dry chemical.

Unsuitable Extinguishing Media: Do not use a heavy water jet. Use of heavy stream of water may spread fire.

Special Hazards Arising From the Substance or Mixture

Fire Hazard: Not flammable.

Explosion Hazard: Product is not explosive.

Reactivity: Thermal decomposition generates: Carbon oxides (CO, CO₂). Nitrogen oxides (NO_x). When heated to decomposition, emits toxic fumes.

Advice for Firefighters

Precautionary Measures Fire: Exercise caution when fighting any chemical fire.

Firefighting Instructions: Use water spray or fog for cooling exposed containers.

Protection During Firefighting: Do not enter fire area without proper protective equipment, including respiratory protection.

Hazardous Combustion Products: Carbon oxides (CO, CO₂). Nitrogen oxides (NO_x).

Other information: Do not allow run-off from fire fighting to enter drains or water courses.

Reference to Other Sections

Refer to section 9 for flammability properties.

SECTION 6: ACCIDENTAL RELEASE MEASURES

Personal Precautions, Protective Equipment and Emergency Procedures

General Measures: Do not get on skin, eyes, or clothing. Do not allow product to spread into the environment.

For Non-Emergency Personnel

Protective Equipment: Use appropriate personal protection equipment (PPE).

Emergency Procedures: Evacuate unnecessary personnel.

For Emergency Personnel

Protective Equipment: Equip cleanup crew with proper protection.

Emergency Procedures: Ventilate area.

Environmental Precautions

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it.

TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Methods and Material for Containment and Cleaning Up

For Containment: Contain any spills with dikes or absorbents to prevent migration and entry into sewers or streams.

Methods for Cleaning Up: Clear up spills immediately and dispose of waste safely. Contact competent authorities after a spill. Collect spillage with sawdust, kieselgur (diatomite), sand, universal binder, Absorbents and binders.

Reference to Other Sections

See heading 8, Exposure Controls and Personal Protection.

SECTION 7: HANDLING AND STORAGE

Precautions for Safe Handling

Additional Hazards When Processed: When heated to decomposition, emits toxic fumes.

Hygiene Measures: Handle in accordance with good industrial hygiene and safety procedures. Wash hands and other exposed areas with mild soap and water before eating, drinking, or smoking and again when leaving work. Do no eat, drink or smoke when using this product. Wash hands and forearms thoroughly after handling.

Conditions for Safe Storage, Including Any Incompatibilities

Technical Measures: Comply with applicable regulations.

Storage Conditions: Store in a dry, cool and well-ventilated place. Keep container closed when not in use. Keep/Store away from extremely low temperatures, incompatible materials.

Incompatible Materials: Strong oxidizers.

Recommended Storage Temperature: 5 – 25°C/41 – 77°F.

Specific End Use(s) Neutral stone prevention detergent. For professional use only.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

Control Parameters

No Occupational Exposure Limits (OELs) have been established for this product or any of its chemical components.

Exposure Controls

Appropriate Engineering Controls: Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas.

Personal Protective Equipment: Protective goggles. Gloves.



Materials for Protective Clothing: N/A.

Hand Protection: Wear chemically resistant protective gloves.

Eye Protection: Chemical goggles or face shield

Skin and Body Protection: None usually needed.

Respiratory Protection: A respirator is not needed under normal and intended conditions of use. In case of inadequate ventilation wear respiratory protection.

Other Information: When using, do not eat, drink or smoke.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on Basic Physical and Chemical Properties

Physical State	: Liquid.
Appearance	: Various.
Odor	: Characteristic.
Odor Threshold	: Not determined.
pH	: 6.5 – 8
Relative Evaporation Rate (butylacetate=1)	: Not determined.
Melting Point	: Not determined.
Freezing Point	: Not determined.
Boiling Point	: 100°C (212°F)
Flash Point	: Not determined
Auto-ignition Temperature	: 400°C (752°F)
Decomposition Temperature	: Not determined.

TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Flammability (solid, gas)	: Not relevant.
Lower Flammable Limit	: Not determined
Upper Flammable Limit	: Not determined
Vapor Pressure	: 32 hPa at 25°C
Relative Vapor Density at 20°C	: Not determined.
Specific Gravity	: 1.18 – 1.28
Solubility	: Not determined.
Partition coefficient: n-octanol/water	: Not determined.
Viscosity	: Not determined.
Explosion Data	: None.
Oxidizing Data	: None.

SECTION 10: STABILITY AND REACTIVITY

Reactivity: Thermal decomposition generates: Carbon oxides (CO, CO₂). Nitrogen oxides (NO_x).

Chemical Stability: Stable under normal conditions.

Possibility of Hazardous Reactions: Hazardous polymerization will not occur.

Conditions to Avoid: Extremely low temperatures. Incompatible materials.

Incompatible Materials: Strong oxidizers.

Hazardous Decomposition Products: Carbon oxides (CO, CO₂). Thermal decomposition generates: Nitrogen oxides.

SECTION 11: TOXICOLOGICAL INFORMATION

Information on Toxicological Effects - Product

Acute Toxicity: Not classified.

LD50 and LC50 Data: Not determined.

Skin Corrosion/Irritation: Not classified.

Serious Eye Damage/Irritation: Not classified.

Respiratory or Skin Sensitization: Not classified.

Germ Cell Mutagenicity: Not classified.

Teratogenicity: Not classified.

Carcinogenicity: Not classified.

Specific Target Organ Toxicity (Repeated Exposure): Not classified.

Reproductive Toxicity: Not classified.

Specific Target Organ Toxicity (Single Exposure): Not classified.

Aspiration Hazard: Not classified.

Symptoms/Injuries After Inhalation: None under normal and intended conditions of use.

Symptoms/Injuries After Skin Contact: None.

Symptoms/Injuries After Eye Contact: None.

Symptoms/Injuries After Ingestion: Ingestion is likely to be harmful or have adverse effects.

Information on Toxicological Effects - Ingredient(s)

LD50 and LC50 Data:

Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
LD50 Oral Rat	940 mg/kg
LD50 Dermal Rabbit	>1650 mg/kg
Zinc sulfate heptahydrate (7446-20-0)	
LD50 Oral Rat	2150 mg/kg
LD50 Intraperitoneal Rat	200 mg/kg

SECTION 12: ECOLOGICAL INFORMATION

Toxicity Not classified.

Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
LC50 Fish 1	368 mg/l (Exposure time: 96h - Species: Rainbow trout)
EC50 Daphnia 1	527 mg/l (Exposure time: 48h - Species: Daphnia magna)
EC50 Algae	9.16 mg/l (Exposure time: 96h - Species: Selenastrum capricornutum)
Zinc sulfate heptahydrate (7446-20-0)	
LC50 Fish 1	1-10 mg/l (Exposure time: 96h - Species: other fish)

TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

Aquatic invertebrate	Not determined
----------------------	----------------

Persistence and Degradability

TM 2000 RB (AFCO 5652)	
Persistence and Degradability	Not established.
Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
Closed bottle BOD/COD 5% - degrades after acclimatization.	

Bioaccumulative Potential

TM 2000 RB (AFCO 5652)	
Bioaccumulative Potential	Not established.
Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
Not expected to bioaccumulate.	

Mobility in Soil

Not available.

Other Adverse Effects

Other Information: Avoid release to the environment.

SECTION 13: DISPOSAL CONSIDERATIONS

Waste Disposal Recommendations: Dispose of waste material in accordance with all local, regional, national, provincial, territorial and international regulations.

SECTION 14: TRANSPORT INFORMATION

14.1 In Accordance with DOT	:	Not regulated for transport.
14.2 In Accordance with IMDG	:	Not regulated for transport.
14.3 In Accordance with IATA	:	Not regulated for transport.
14.4 In Accordance with TDG	:	Not regulated for transport.

SECTION 15: REGULATORY INFORMATION

US Federal Regulations

TM 2000 RB (AFCO 5652)	
2-phosphonobutane 1,2, 4 tricarboxylic acid, tetrasodium salt (6666-53-2)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	
Zinc sulfate heptahydrate (7446-20-0)	
Listed on the United States TSCA (Toxic Substances Control Act) inventory.	

US State Regulations: No components listed.

Canadian Regulations

Sodium salt of hydroxyethylidene-1,1, diphosphonic acid (29329-71-3)	
Listed on the Canadian DSL (Domestic Substances List) inventory.	
2-phosphonobutane 1,2, 4 tricarboxylic acid, tetrasodium salt (6666-53-2)	
Listed on the Canadian DSL (Domestic Substances List) inventory.	
Zinc sulfate heptahydrate (7446-20-0)	
Listed on the Canadian DSL (Domestic Substances List) inventory.	

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

Revision date	:	03/10/2025
Other Information	:	This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200.

TM 2000 RB (AFCO 5652)

Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations

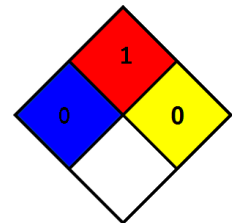
GHS Full Text Phrases:

Acute Tox. 4 (Oral)	Acute toxicity (oral) Category 3.
Aquatic Acute 1	Hazardous to the aquatic environment - Acute Hazard Category 1.
Aquatic Chronic 1	Hazardous to the aquatic environment - Chronic Hazard Category 1.
Eye Dam. 1	Serious eye damage/eye irritation Category 1.
Eye Irrit. 2A	Serious eye damage/eye irritation Category 2A.
H302	Harmful if swallowed.
H318	Causes serious eye damage.
H319	Causes serious eye irritation.
H402	Harmful to aquatic life.
H412	Harmful to aquatic life with long lasting effects.

NFPA Health Hazard : 0 - Exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials.

NFPA Fire Hazard : 1 - Must be preheated before ignition can occur.

NFPA Reactivity : 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.



HMIS III Rating

Health : 0 - Minimal Hazard - No significant risk to health.

Flammability : 1 - Slight Hazard.

Physical : 0 - Minimal Hazard.

Party Responsible for the Preparation of This Document

AFCO
550 Development Avenue
Chambersburg, PA 17201
T: 800-345-1329

This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.

North America GHS SDS 2015 (U.S., Can., Mex.)